

Moving Charges & Magnetism

#Electromagnetic Force on moving charge:

→ Force on moving charge in presence of both $\vec{E}$ & $\vec{B}$ given by 'Lorentz eqn'

$\vec{F}_{\text{net}} = q\vec{E} + q(\vec{v} \times \vec{B})$

$\vec{F}_{\text{net}}$ is absolute force & frame independent.

While, magnetic & electric forces are frame dependent.

→ For dirⁿ of magnetic force use RHTA.

#Magnetic Force on Moving Charge:

$$\vec{F}_m = q(\vec{v} \times \vec{B})$$

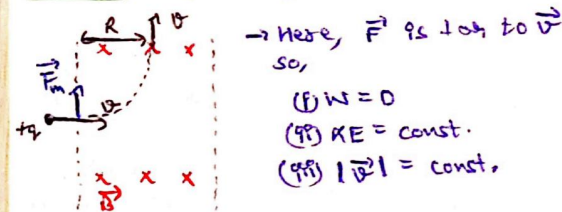
→ For force dirⁿ, we can also use right hand palm rule.

Finger $\Rightarrow \vec{B}$

Thumb $\Rightarrow \vec{v}$

Palm $\Rightarrow$ gives $\vec{F}$ (In same dirⁿ for +q, In opp dirⁿ for -q)

Case-01: $(\vec{v} \perp \vec{B})$

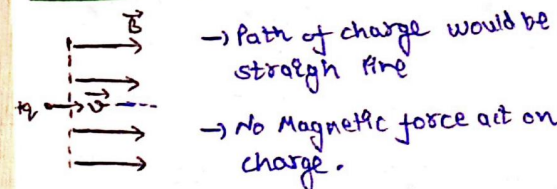


→ Path of charge would be circular.

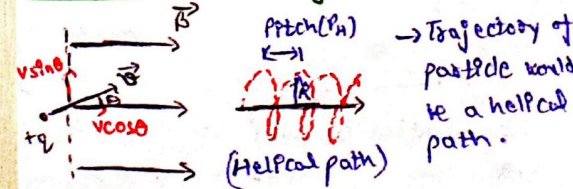
$$R = \frac{mv}{qB} \quad \omega = \frac{qB}{m} \quad T = \frac{2\pi m}{qB}$$

($qB = \frac{mv^2}{R}$) ($\omega = \frac{v}{R}$) ($T = \frac{2\pi}{\omega}$)

Case-02: $(\vec{v} \parallel \vec{B})$



Case-03: $(\vec{v}$ at angle θ to $\vec{B}$)



Here

$$F = qvB \sin \theta$$

$$T = \frac{2\pi m}{qB}$$

→ For a helix, we calculate two of its parameters.

$$R_H = \frac{mv \sin \theta}{qB} \quad (\text{Radius of Helix})$$

$$P_n = v \cos \theta \times T = \frac{2\pi m v \cos \theta}{qB}$$

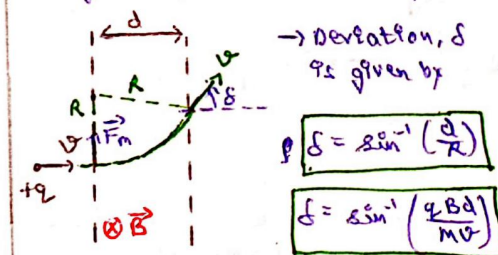
(Pitch of Helix)

NOTE: For time period of a charge doing circular motion under uniform $\vec{B}$.

- Independent of velocity
- Depends only on mass to charge ratio of particle

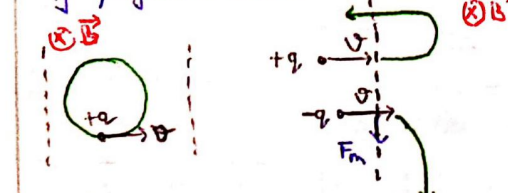
#Deflection by a Sector of MF:

→ The particle moves on a circular path in MF region, and exits it tangentially to the circular path.

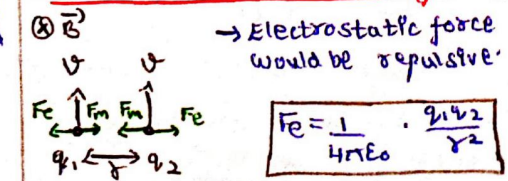


(If $R < d$, then particle will complete a semi-circle & $\delta = \pi$)

→ Particle will complete circle only if projected entire in MF.



#Force b/w two parallel moving charge:



→ Magnetic force would apply in attractive fashion.

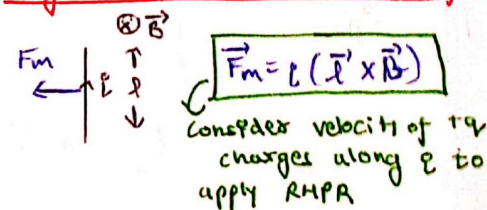
Magnetic force on q_2 due to q_1

$$F_m = q_2 \cdot v \cdot B_1$$

$$\Rightarrow F_m = \frac{\mu_0}{4\pi} \frac{q_1 q_2 v^2}{r^2}$$

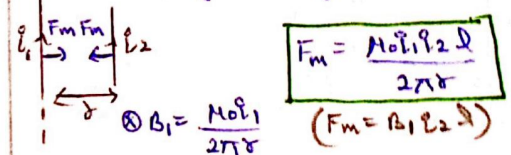
(If $v = c$ (speed of light), then $F_e = F_m$)

#Magnetic Force on Current Carrying wire:



#Force b/w parallel Current Carrying Wire:

→ One of the wires must be of ∞ length

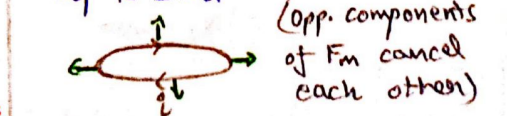


→ Current in same $\Rightarrow \vec{F}_m$ is attractive direction

Current in opp $\Rightarrow \vec{F}_m$ is repulsive direction

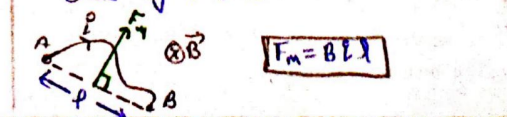
→ Due to above phenomenon, when current is passed through a conducting spring, it compresses.

→ Therefore, magnetic force on a closed loop is zero.

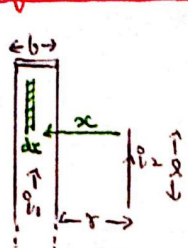


#Magnetic Force on Random shaped wire:

→ Take 'q' as distance b/w both ends of the wire.



Magnetic Force b/w a parallel wire & strip:



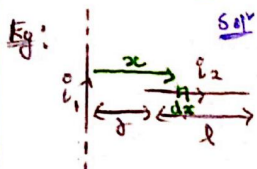
$$dI = \frac{i_2}{b} \cdot dx$$

Force on i_2 due to i_1

$$dF = \frac{\mu_0 i_1 i_2 \cdot dx \cdot l}{4\pi r^2}$$

$$F = \frac{\mu_0 i_1 i_2 l}{4\pi b} \ln\left(\frac{r+b}{r}\right)$$

$$\Rightarrow F = \frac{\mu_0 i_1 i_2 l}{4\pi b} \ln\left(\frac{r+b}{r}\right)$$

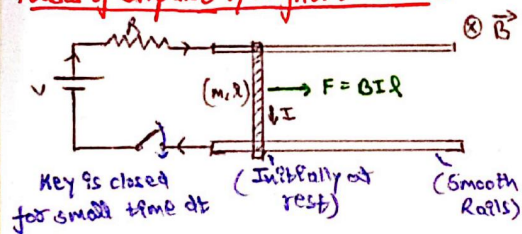


$$dF = B \cdot i_2 \cdot dx$$

$$F = \int_0^l \frac{\mu_0 i_1 i_2}{2\pi r} \cdot dx$$

$$\Rightarrow F = \frac{\mu_0 i_1 i_2 l}{2\pi} \ln\left(\frac{r+b}{r}\right)$$

Cases of Impulse by Magnetic Force:



→ When current passes through conductor, it experiences magnetic force & starts moving.

$$Q = I \cdot dt$$

Charge passed through conductor in dt time

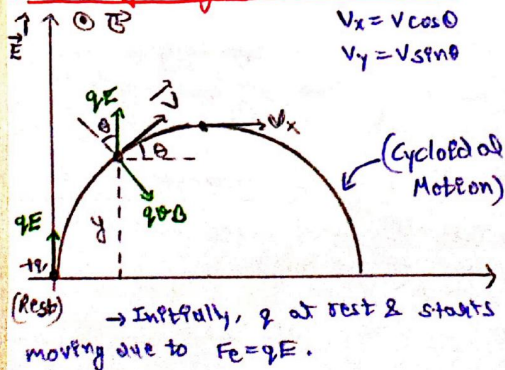
→ If conductor attains v velocity after dt,

$$BIl \cdot dt = mv - 0$$

$$\Rightarrow J = Bql = mv$$

(Impulse due to F_m) ($\because q = I \cdot dt$)

Motion of charge in both EF & MF:



$$v_x = v \cos \theta$$

$$v_y = v \sin \theta$$

(Cycloid Motion)

→ Initially, q at rest & starts moving due to $F_e = qE$.

→ As q moves, F_m also starts acting on it, always $\perp$ to its velocity.

→ At some height y

(i) Work done by F_e will convert into KE (as F_m work = 0)

$$qEy = \frac{1}{2}mv^2$$

$$\Rightarrow v = \sqrt{\frac{2qEy}{m}}$$

Velocity at any height y

(ii) Radius of curvature can be calculated by net centripetal force.

$$qvB - qE \cos \theta = \frac{mv^2}{r}$$

v → velocity at that instant

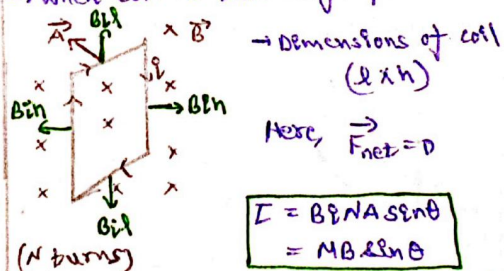
r → Radius of curvature

θ → Angle made by net v with horizon.

Torque on a coil in Magnetic Field:

→ When coil is in plane of $\vec{B}$, no torque & $F_{net} = 0$

→ When coil is tilted slightly,



→ Dimensions of coil ($l \times h$)

Here, $F_{net} = 0$

$$\tau = BINA \sin \theta$$

$$= MB \sin \theta$$

$$\vec{M} = I \vec{A} \times \vec{N}$$

M → Magnetic moment ($N \cdot m$)

θ → Angle b/w $\vec{A}$ & $\vec{B}$

→ Vectorially,

$$\vec{\tau} = \vec{M} \times \vec{B}$$

Interaction Energy of coil in MF & its Equilibrium:

→ In above case, the potential energy of coil is given by

$$U = -\vec{M} \cdot \vec{B} = -MB \cos \theta$$

$$U_{max} = MB \quad (\theta = 180^\circ) \text{ Unstable eqm}$$

$$U_{min} = -MB \quad (\theta = 0^\circ) \text{ Stable eqm}$$

→ Work done in rotating/changing orientation of coil in MF

$$W_{ext} = U_2 - U_1 = MB(\cos \theta_1 - \cos \theta_2)$$

Magnetic flux through a current carrying coil:

$$\Phi_B = \vec{B} \cdot \vec{A} \quad (\text{if } \vec{B} \text{ is uniform})$$

$$\Phi_B = \int \vec{B} \cdot d\vec{A} \quad (\text{if } \vec{B} \text{ is variable})$$

→ Since Magnetic field always form closed loops, for a closed area

$$\oint \vec{B} \cdot d\vec{A} = 0 \quad (\text{Always})$$

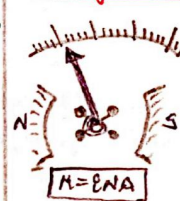
→ Work done in rotating coil in MF

$$W_{ext} = -\Delta \Phi_B$$

I → Current flowing in coil

$$\Delta \Phi_B \rightarrow (\Phi_B)_f - (\Phi_B)_i$$

Moving Coil Galvanometer:



→ Cylindrical magnet taken to produce almost straight MF.

→ $\vec{B}$ is always $\perp$ to coil, no matter how much it rotates.

$$\therefore \tau = MB \sin 90^\circ = ENAB$$

$$\tau = C\theta = ENAB \quad (C \rightarrow \text{Restoring torque})$$

$$\theta = \left(\frac{C}{NAB}\right) \tau \quad (C \rightarrow \text{Torsional const. of spring})$$

$$\text{Current sensitivity of Galvanometer} = \left(\frac{NAB}{C}\right) \quad (\text{rad/Ampere})$$

Relation b/w $\vec{M}$ & L of a Rotating charge:



We can write,

$$I = q \times \text{rev. freq.}$$

$$I = \frac{q\omega}{2\pi}$$

$$\Rightarrow \text{Magnetic moment } (M) = I \cdot \pi r^2 = \frac{q\omega r^2}{2}$$

$$\text{Now, } \frac{M}{L} = \frac{q}{2m} \quad \left[\begin{array}{l} M \rightarrow \text{Mag. moment} \\ L \rightarrow \text{Angular momentum} \end{array} \right]$$

This relation is valid for every rigid body whose mass density & charge density, both are uniform

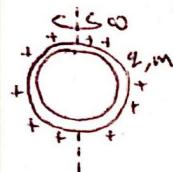
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→ So, for a rotating charge

$$\frac{M}{L} = \frac{q}{2m} \Rightarrow \frac{q\omega r^2}{2L} = \frac{q}{2m}$$

$$\Rightarrow \boxed{L = m\omega r^2}$$

Ex: Hollow sphere rotating with ω .
Find M .



Soln $M = \frac{q}{2m} \cdot L$

$$M = \frac{q}{2m} \cdot I\omega \quad [I \rightarrow MOI]$$

$$\Rightarrow M = \frac{q}{2m} \cdot \frac{2}{3} m r^2 \cdot \omega$$

$$\Rightarrow \boxed{M = \frac{q\omega r^2}{3}} \quad \underline{\text{Ans}}$$

Torque on a 3D loop in Magnetic Field:

→ In case of loops, torque is given by

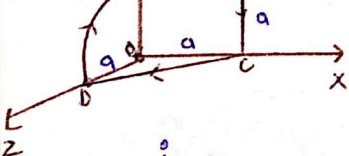
$$[\vec{\tau} = \vec{m} \times \vec{B}]$$

→ Divide the 3D loop into different 2-D loops and assume extra wire to complete those 2-D loops.

→ The extra wires will automatically nullify each other's effect.

$$\vec{B} = B_x \hat{i} + B_y \hat{j} + B_z \hat{k}$$

Find net torque on loop.



Soln

$$M_{ABEO} = i a^2 (-\hat{k})$$

$$M_{BOEC} = \frac{1}{2} i a^2 (-\hat{j})$$

$$M_{DOA} = i \left(\frac{\pi a^2}{4} \right) (-\hat{i})$$

By these M , we can calculate torque.

NOTE: A freely placed loop starts rotating about an axis of rotation ~~at~~ which passes through its C.O.M & along the dirⁿ of $\vec{B}$.